Digital Logic Design

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Course Information

- Course: CS-103 Digital Logic Design
- Instructor: Dr. Muhammad Usman
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- Recommended Books:
  - M. Morris Mano and Charles R. Kime, ***Logic and Computer Design Fundamentals***, Pearson Education Inc.
  - M. Morris Mano, ***Digital Design***, Pearson Education Inc.,
  - Samir Palnitkar, ***Verilog HDL: A Guide to Digital Design and Synthesis,*** Prentice Hall Publisher
  - Albert P. Malvino, Jerald A. Brown, ***Digital Computer Electronics*****,** McGraw-Hill International Editions
  - John F. Wakerly, Digital Design: Principles and Practices

Course Contents

- Number Systems
- Boolean Algebra & Logic Gates
- Combinational Logic
- Simplification
- Sequential Circuits
- Registers & Counters
- Programmable Logic Devices
- Introduction to Memories

Convention

- Every class
  - First 5-10 minutes, review of the last class
- Students will be randomly picked to answer questions
- Class participation is very important

Grading Information

- Grade determinates
  - Midterm 25%
  - Final Exam 50%
  - Homework Assignments 07%
  - In-class pop quizzes 08%
  - Project 10%
- Specially for re-registered students
  - If there is a conflict in timings with other class(es), please sort it out yourself with the semester coordinator.
  - For this course you have to be regular.

Grading Policies

- Assignments should be submitted sharply by the due date.
- No late assignments will be accepted.
- All the assignments should be completed individually. Duplicate assignments will receive duplicate grades of zero. Second offense will result in more strict action.

Course Learning Outcomes (CLOs)

1. Acquire knowledge related to the concepts for the design of digital circuits	PLO-2
2. Apply the acquired knowledge for simplification of logic circuits/expressions	PLO-3
3. Design combinational and sequential circuits	PLO-4
4. Analyze combinational and sequential circuits	PLO-3

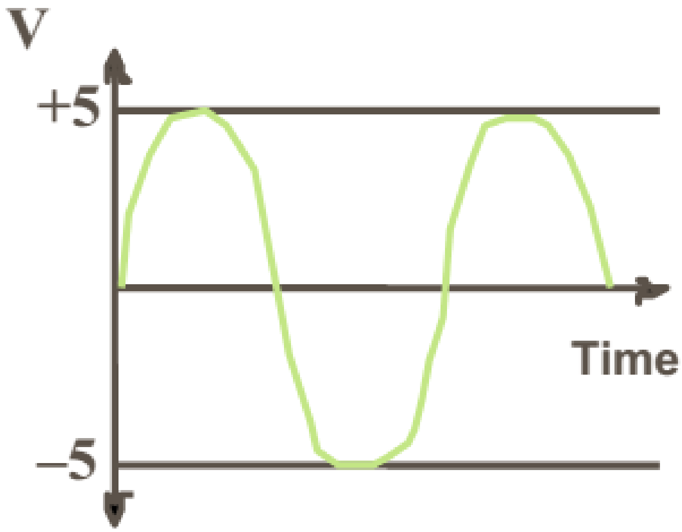
Program Learning Outcomes (PLOs)	**Computing Professional Graduate**
1. Academic Education	To prepare graduates as computing professionals
2. Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3. Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
4. Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
5. Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

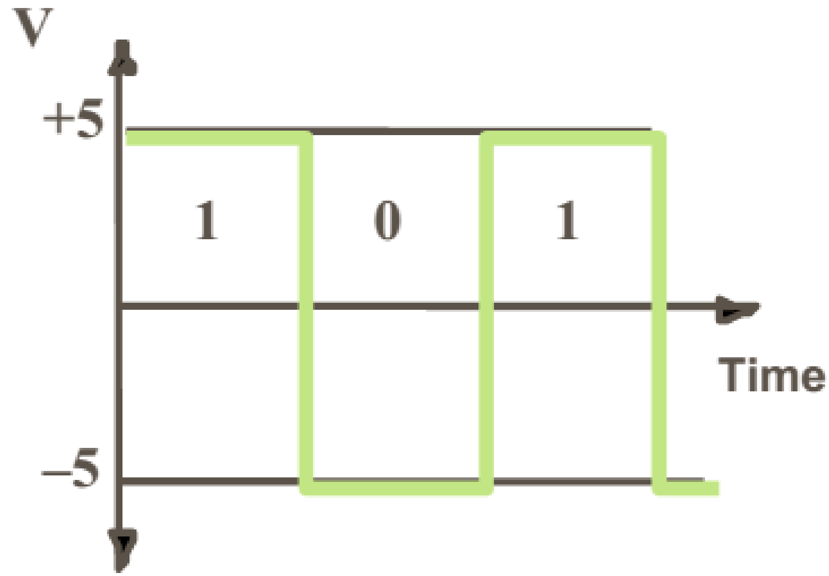
6. Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings
7. Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
8. Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice
9. Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
10. Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

Outline of today's Lecture

- Digital vs Analog
- Technology Trends
- Classes of Computers
- Parts of Computer

Analog vs Digital

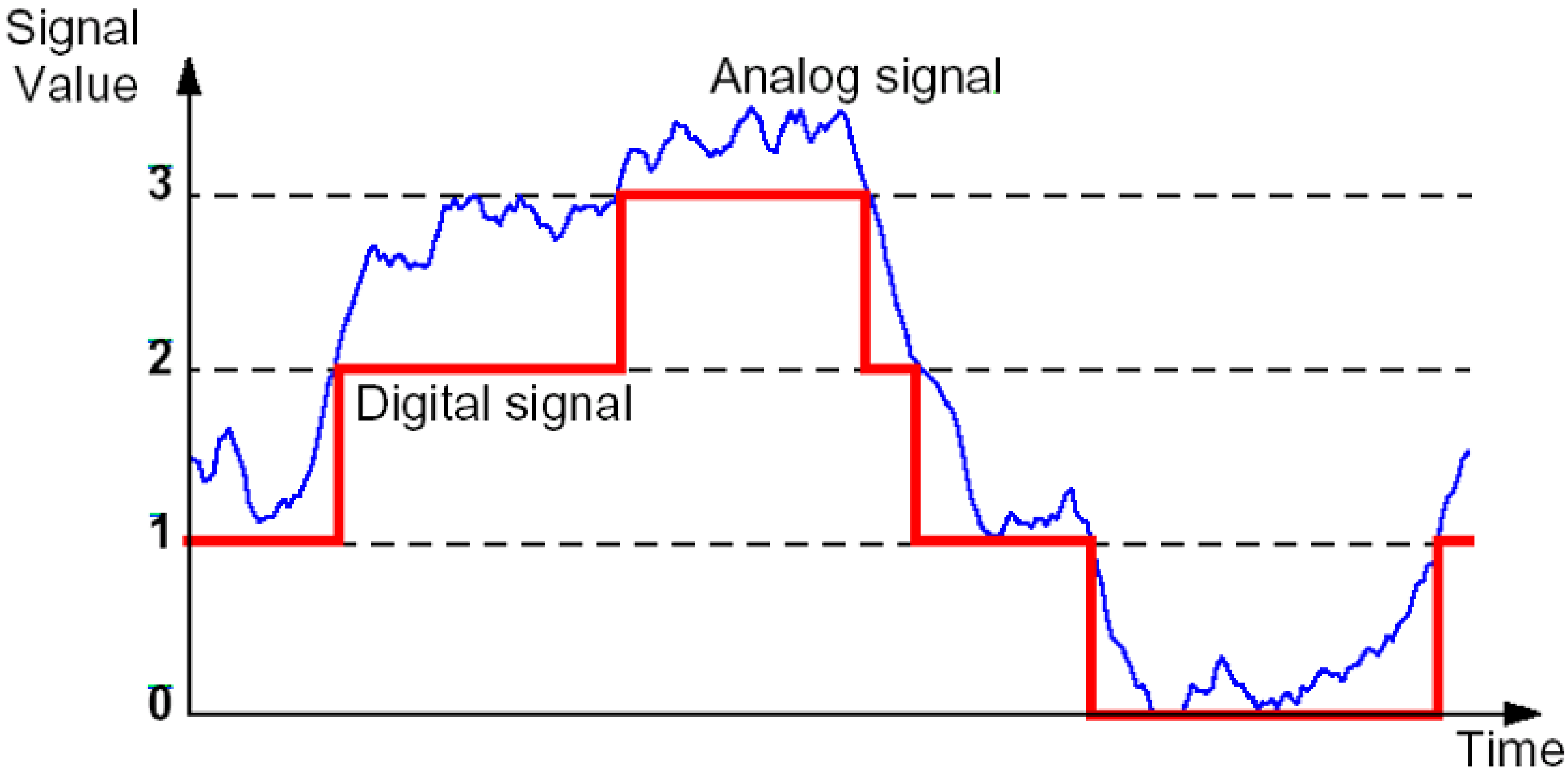




Analog: values vary over a broad range continuously

Digital: only discrete values

Analog Signal vs Digital



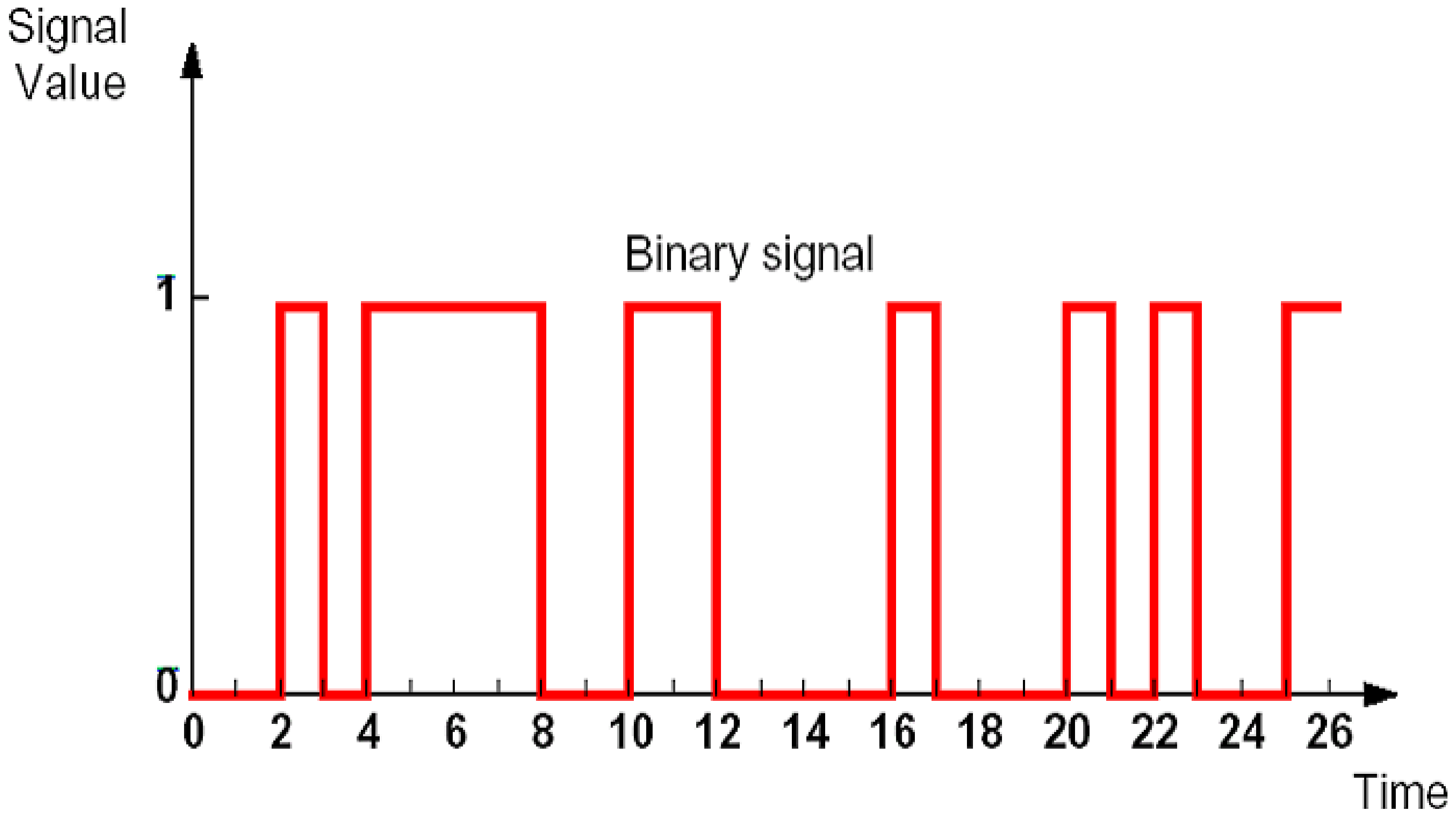
Analog vs Digital

- Analog devices process signal that can assume any value across a continuous range and produce results that are also in continuous form.
- Digital devices process signals that take on only two discrete values such as 0 and 1 and produce output that can be represented by 0 and 1
- Examples
  - Analog Devices: solid-state devices TV (except for digital TV), Audio amplifier etc.
  - Digital Devices: Computer, CD player, digital TV, digital cellular phone, electronic calculator, and digital camera.

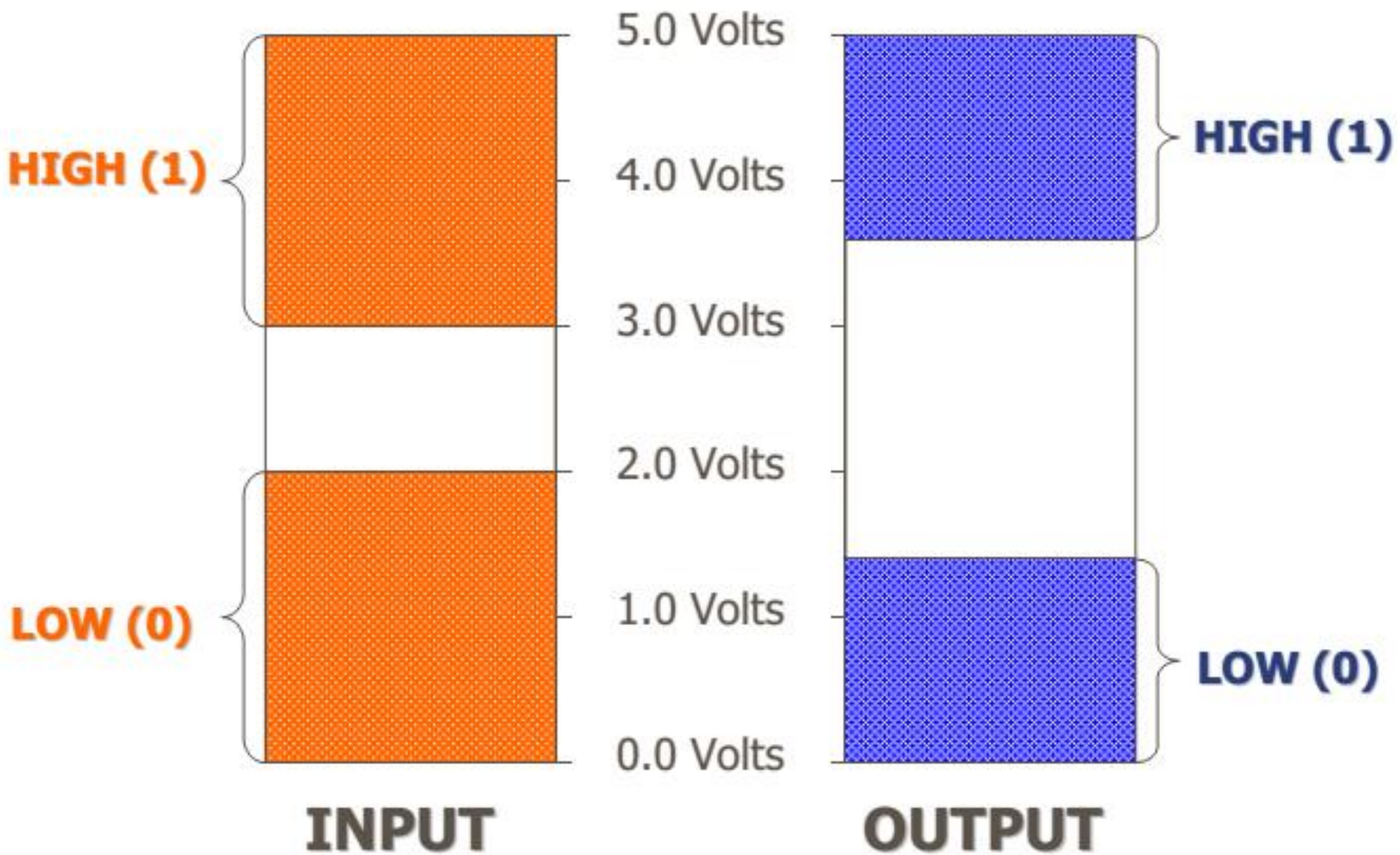
Analog vs. Digital Systems

- Analog systems
  - Limited precision, errors accumulate,
  - Interface circuits (i.e., sensors & actuators) often analog
- Digital systems:
  - More accurate and reliable
  - Readily available as self-contained, easy to cascade building blocks
  - Computers use digital circuits internally

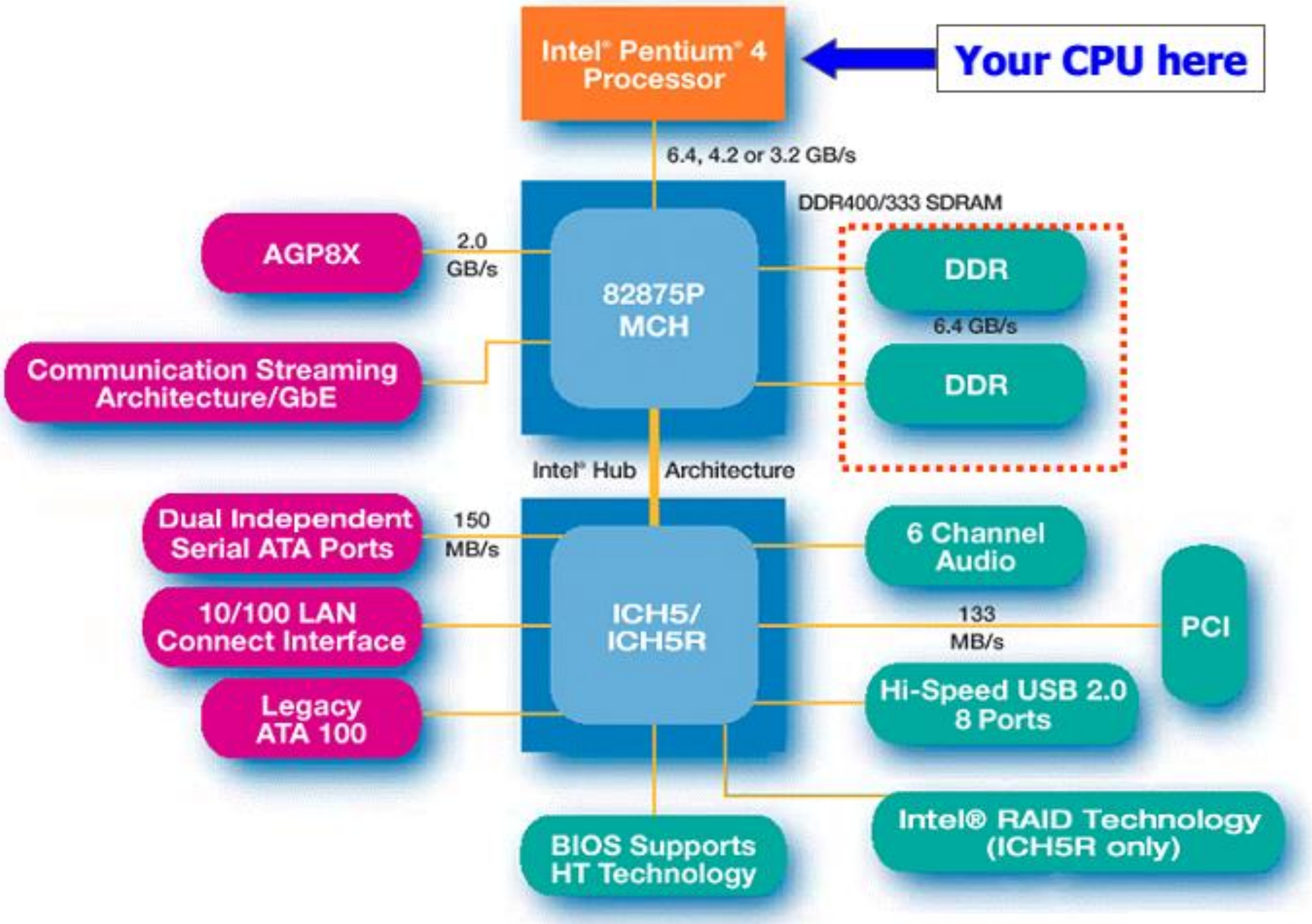
Binary Signal



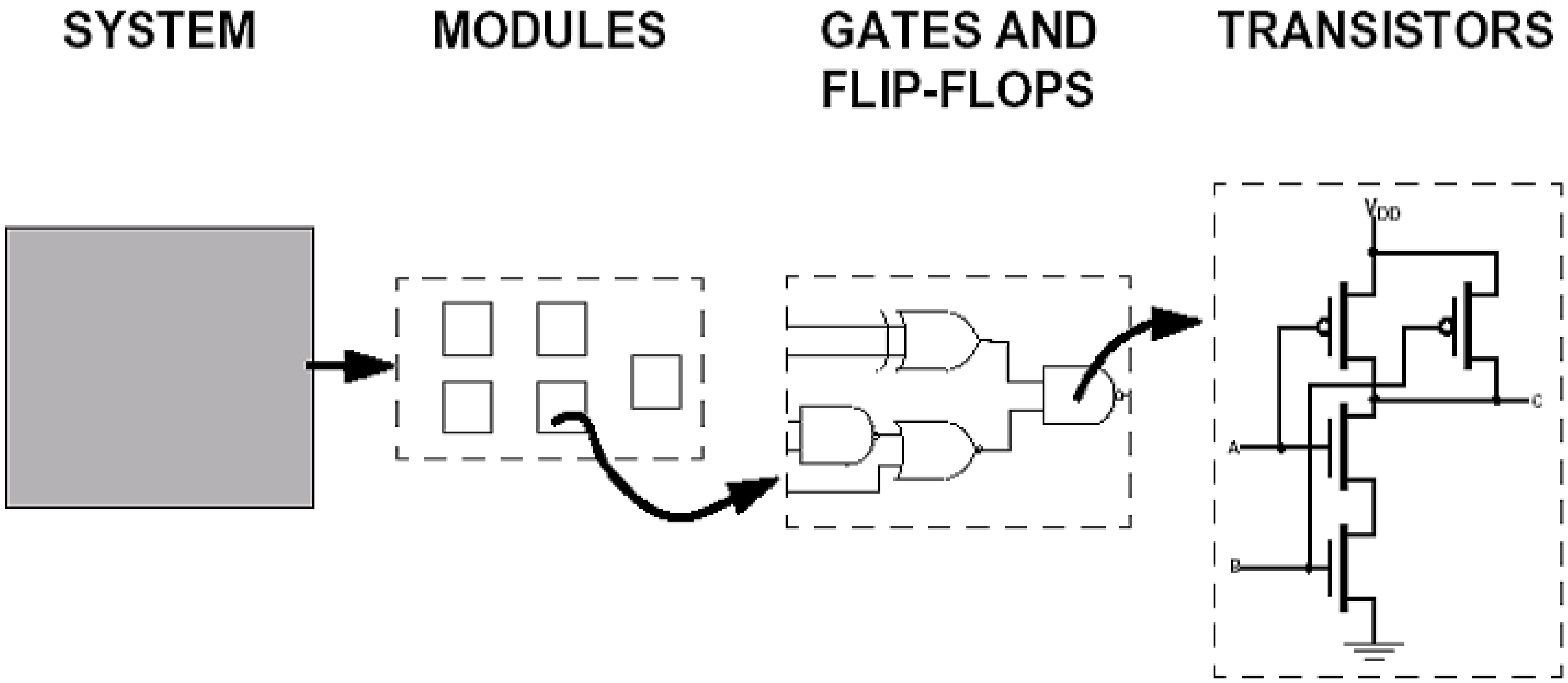
Voltage Range of Binary Signals



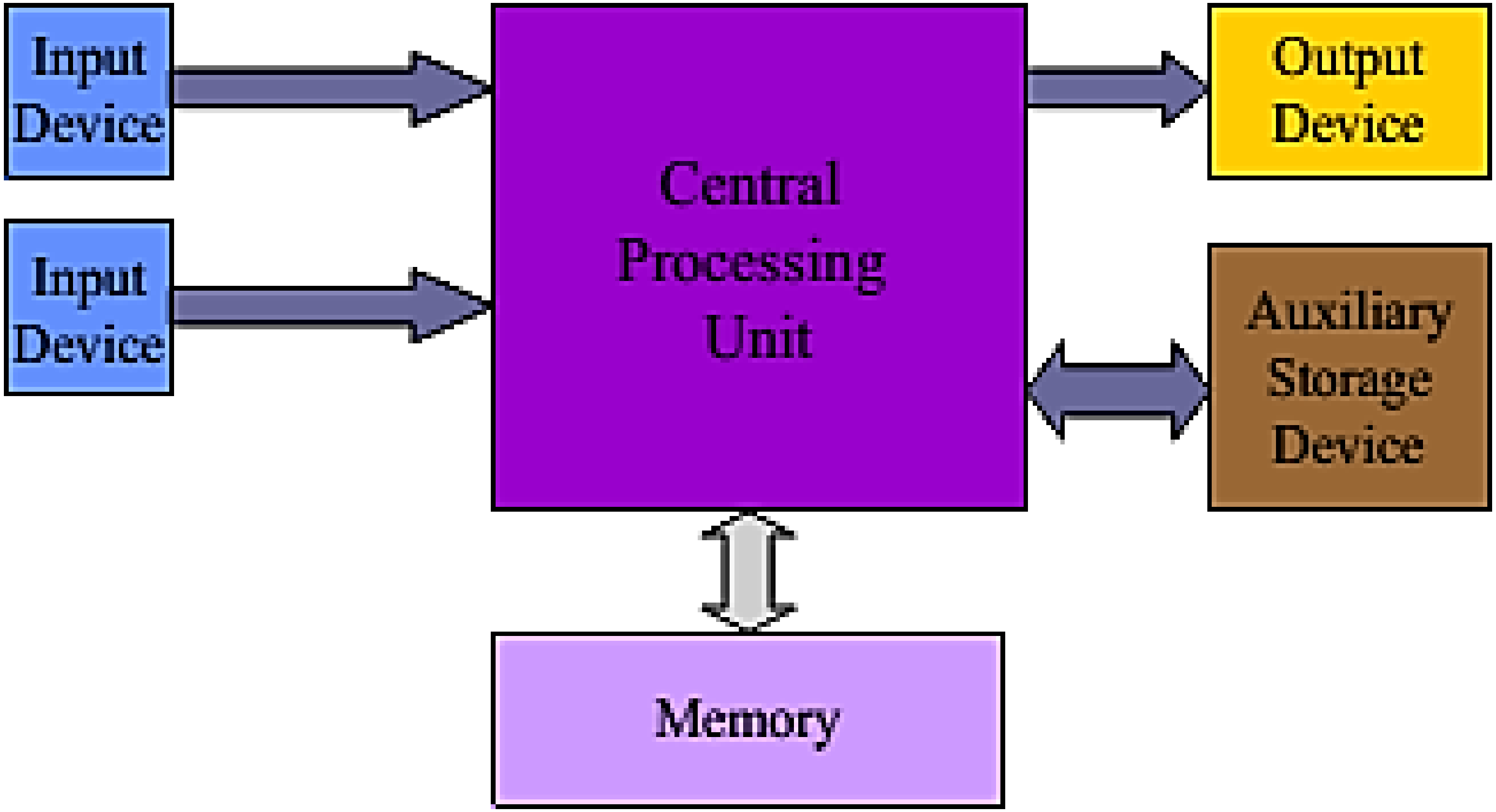
A Generic Intel-based PC System



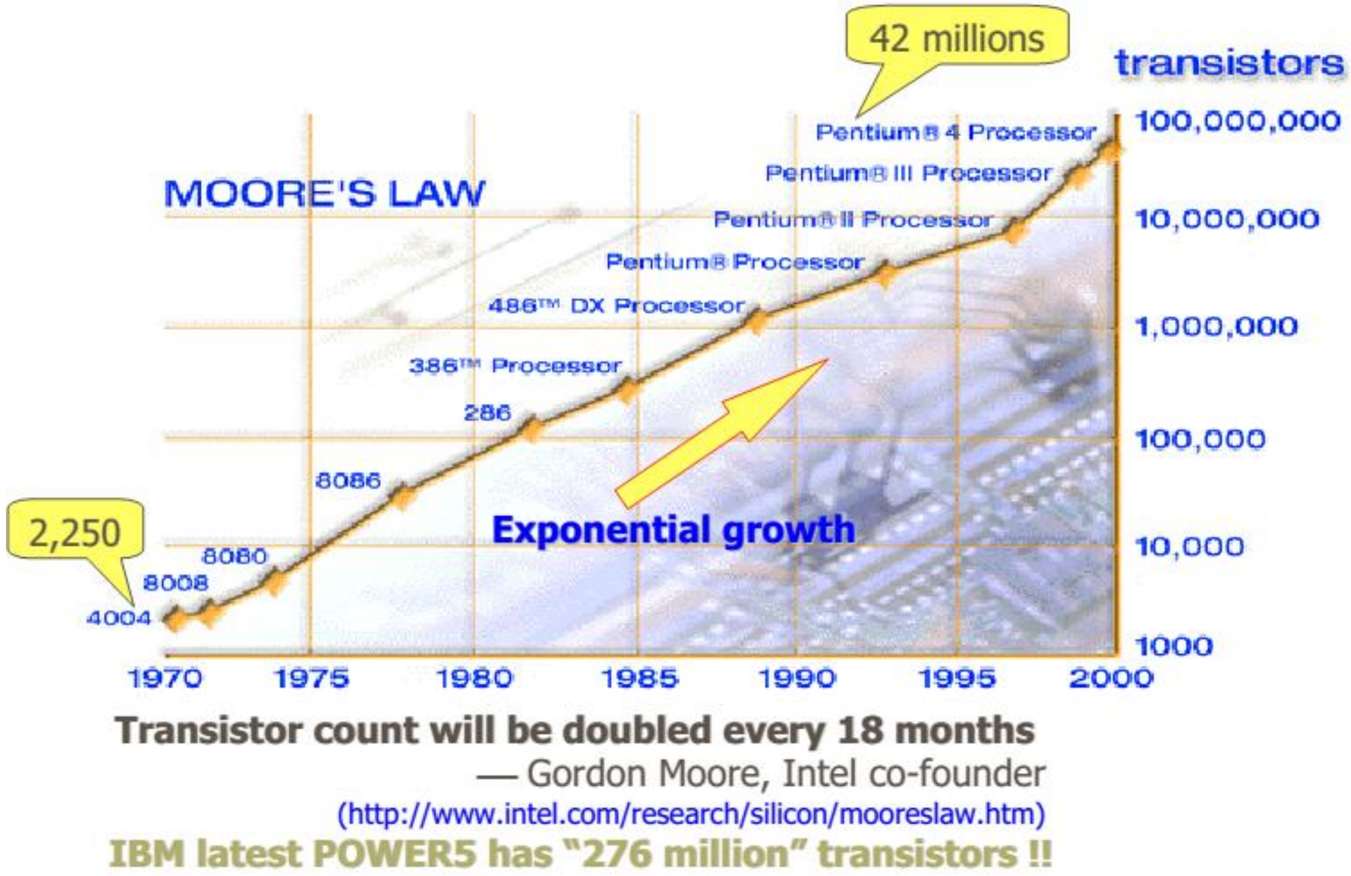
Zoom-in a System Component

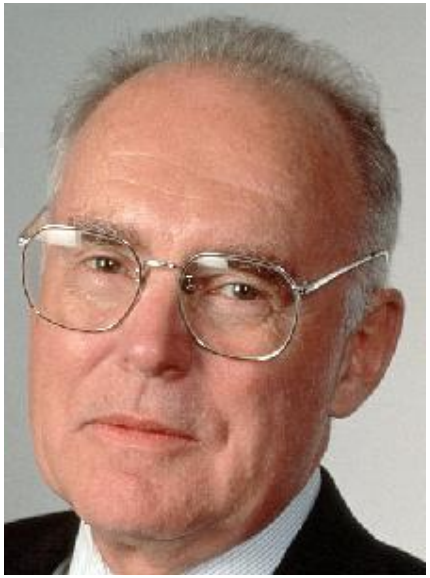


A Computer


Input Device
Input Device
Central Processing Unit
Output Device
Auxiliary Storage Device
Memory

Technology Trends



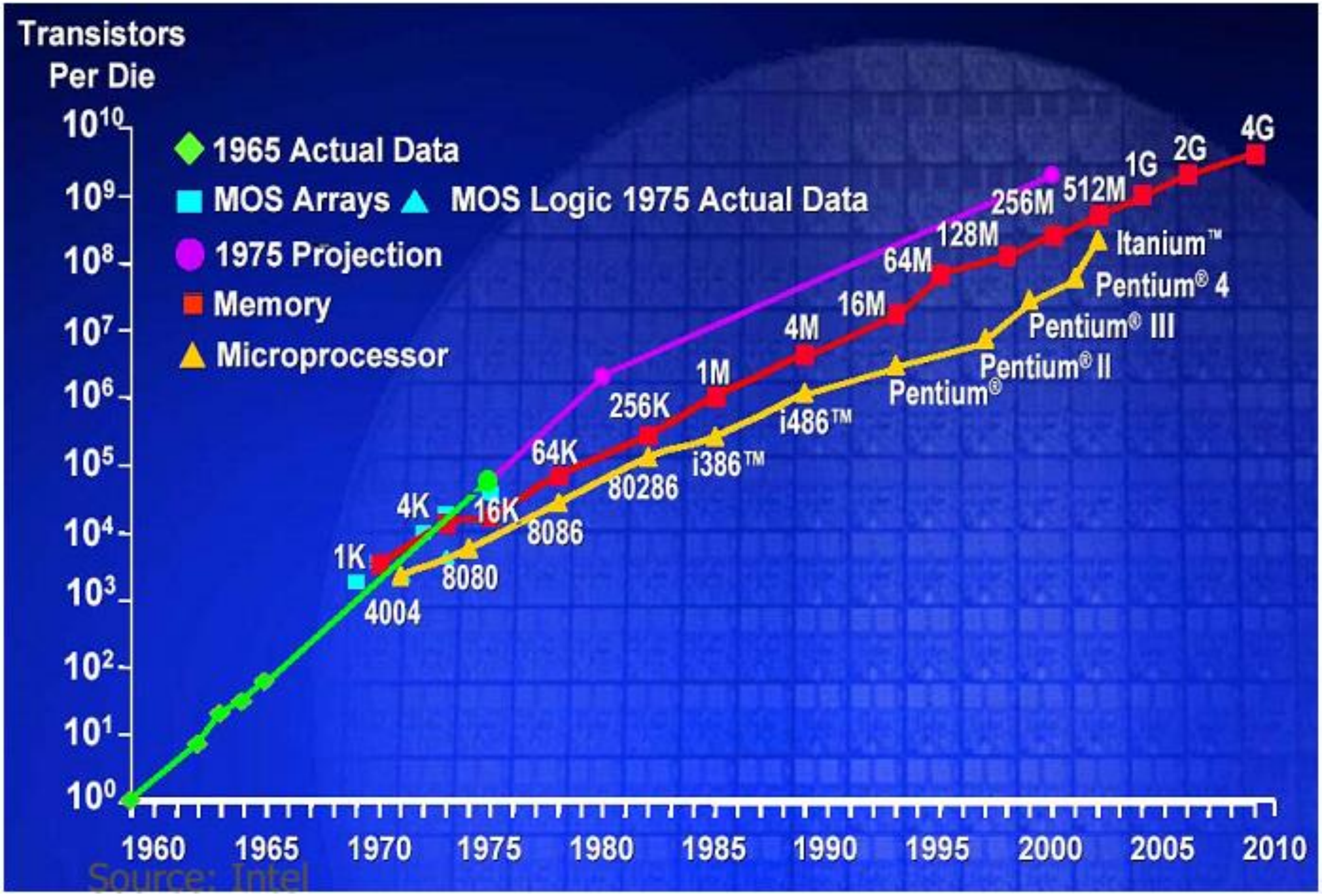


Gordon Moore
Intel Cofounder
B.S. Cal 1950!

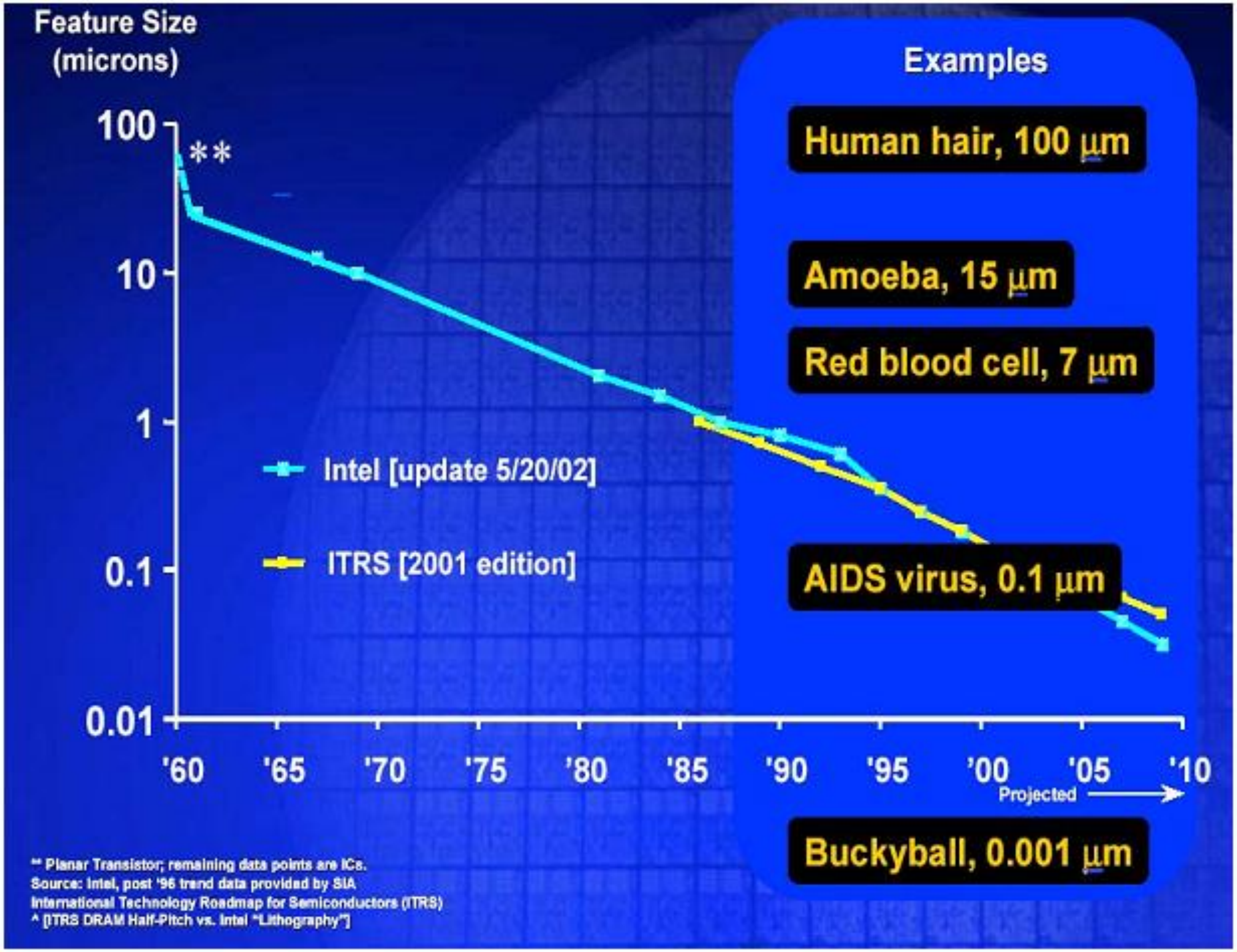
2XTransistors / Chip every 1.5 years

Called "Moore's Law"

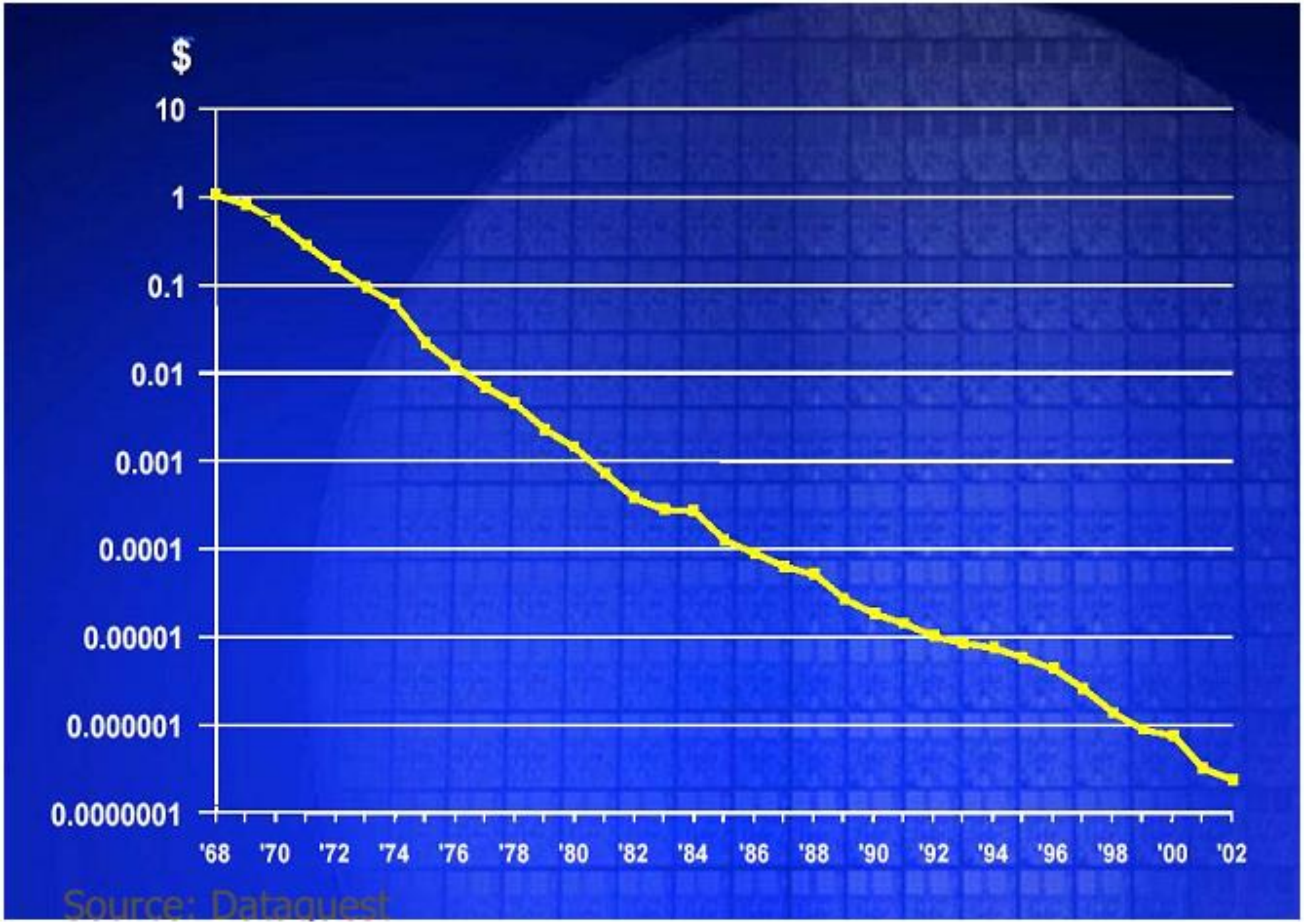
Integrated Circuit Density



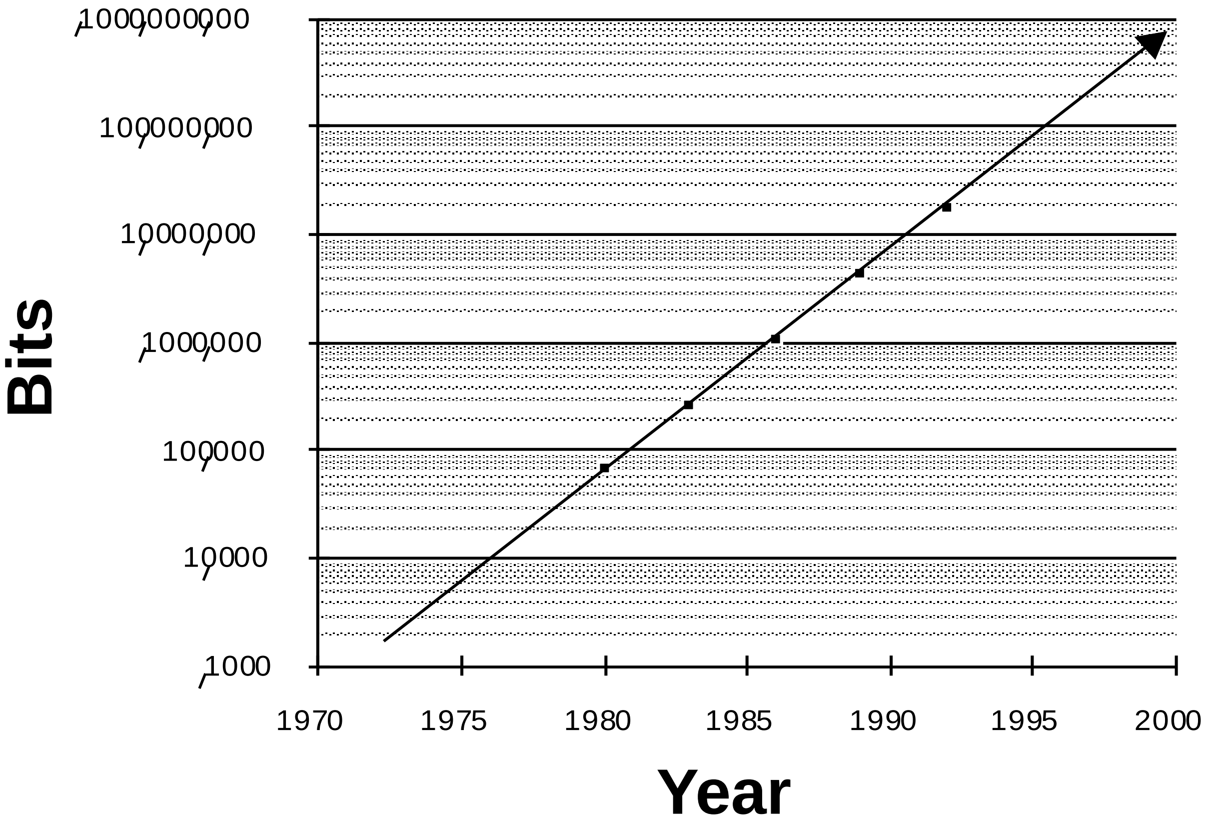
Minimum Feature Size



Avg Transistor Price per year



Memory Capacity (Single-Chip DRAM)



year	size (Mbit)
1980	0.0625
1983	0.25
1986	1
1989	4
1992	16
1996	64
1998	128
2000	256
2002	512
2004	1024 (1Gbit)

- Now 1.4X/yr, or 2X every 2 years.
- 8000X since 1980!

Computer Technology - Dramatic Change!

- Memory
  - DRAM capacity: 2x / 2 years (since ‘96);
    64x size improvement in last decade.
- Processor
  - Speed 2x / 1.5 years (since ‘85);
    100X performance in last decade.
- Disk
  - Capacity: 2x / 1 year (since ‘97)
    250X size in last decade.
- Moore’s Law enables processor
  (2X transistors/chip ~1.5-2 yrs)

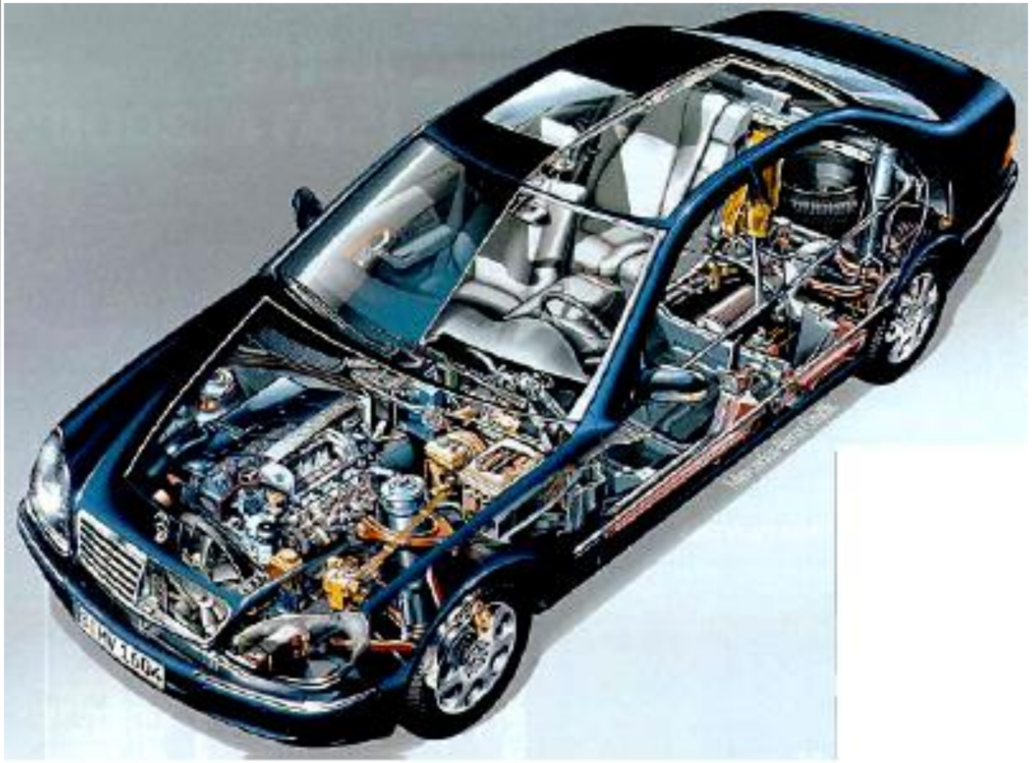
Computer Types

- Desktop/Laptop
  - Most widely used in everyday life

- Workstation
  - Same dimensions as desktop computers
  - High-resolution graphics I/O capability, more computational power

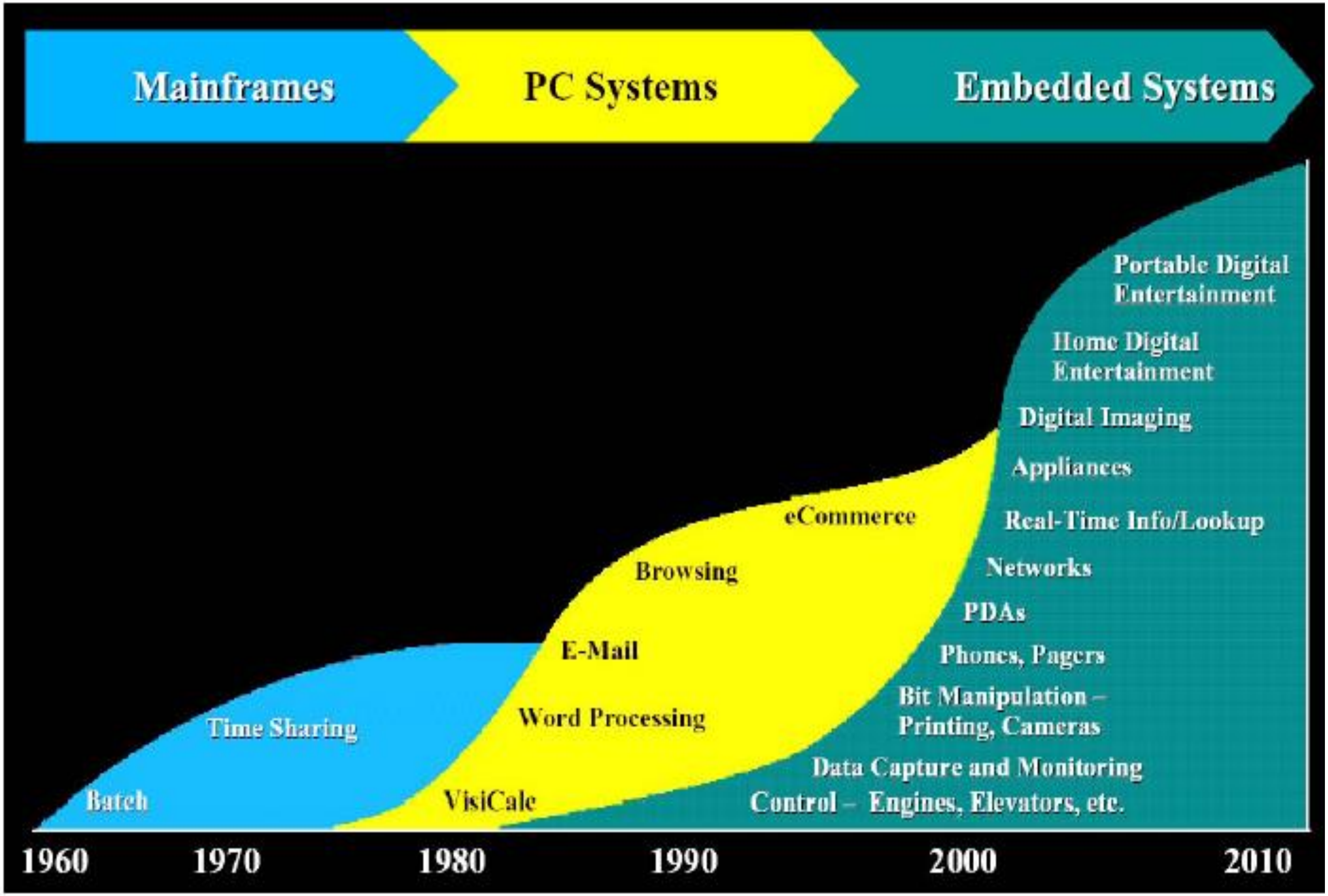
- Servers ~ Supercomputers
  - HP Integrity Superdome, IBM eSer
  - Computing power and storage

- Embedded Computers



- An **embedded** system is a **computer** system with a dedicated function within a larger mechanical or electrical system, often with real-time **computing** constraints.
- It is **embedded** as part of a complete device often including hardware and mechanical parts.

- The micro-controller is embedded in the appliance, you often are not aware of the fact that it contains a micro-controller (e.g. 70 micro-controllers in a modern high end car: engine control, ABS, airbag, interior illumination, central lock, alarm, radio, ...)

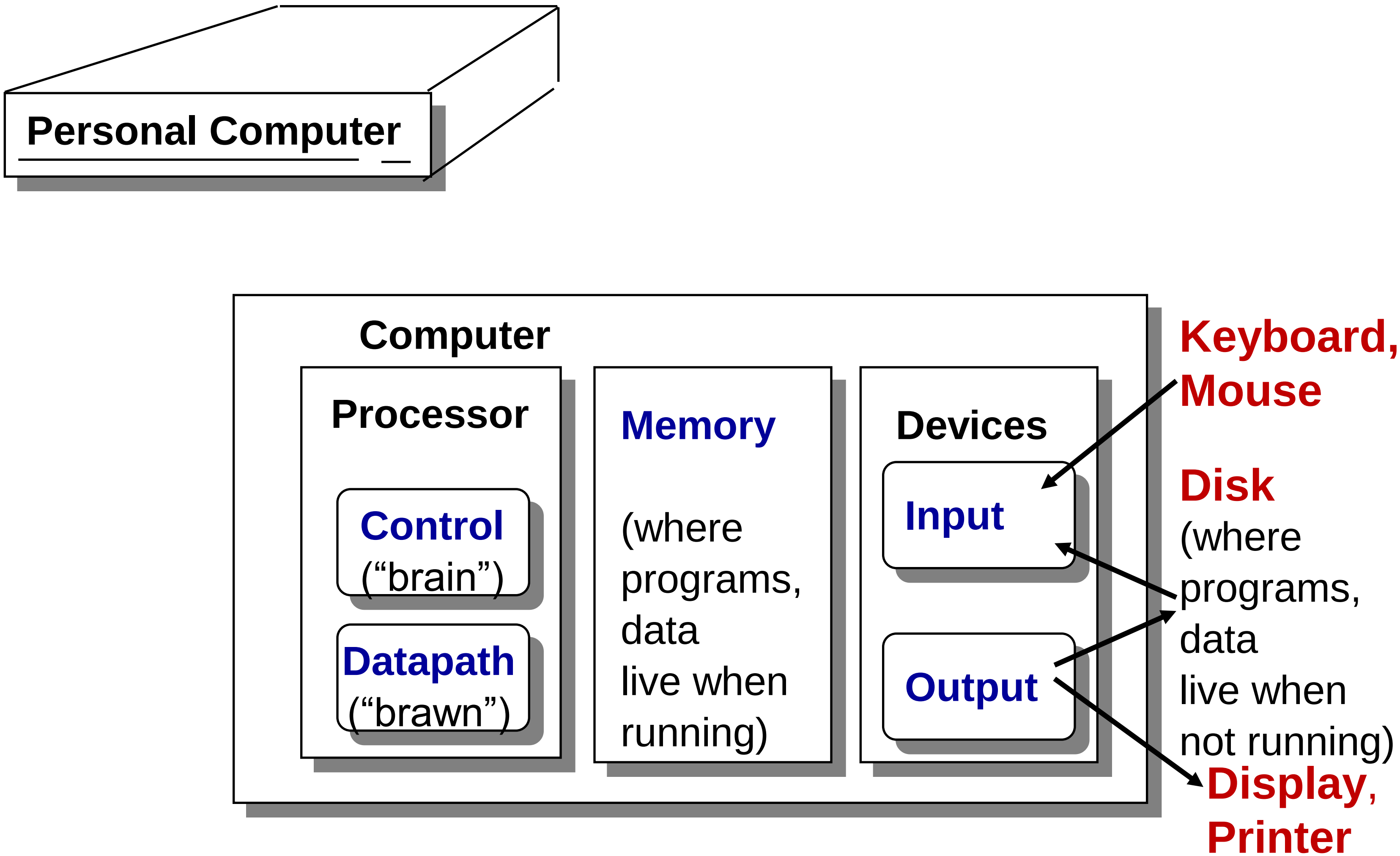

Mainframes
PC Systems
Embedded Systems
Portable Digital Entertainment
Home Digital Entertainment
Digital Imaging
Appliances
eCommerce
Real-Time Info/Lookup
Browsing
Networks
PDAs
E-Mail
Phones, Pagers
Bit Manipulation – Printing, Cameras
Time Sharing
Word Processing
Data Capture and Monitoring
Batch
VisiCalc
Control – Engines, Elevators, etc.
1960
1970
1980
1990
2000
2010

Processor Categories

- Popular processor designs can be broadly divided into two categories:
  - Complex Instruction Set Computer (CISC)
  - Reduced Instruction Set Computers (RISC)

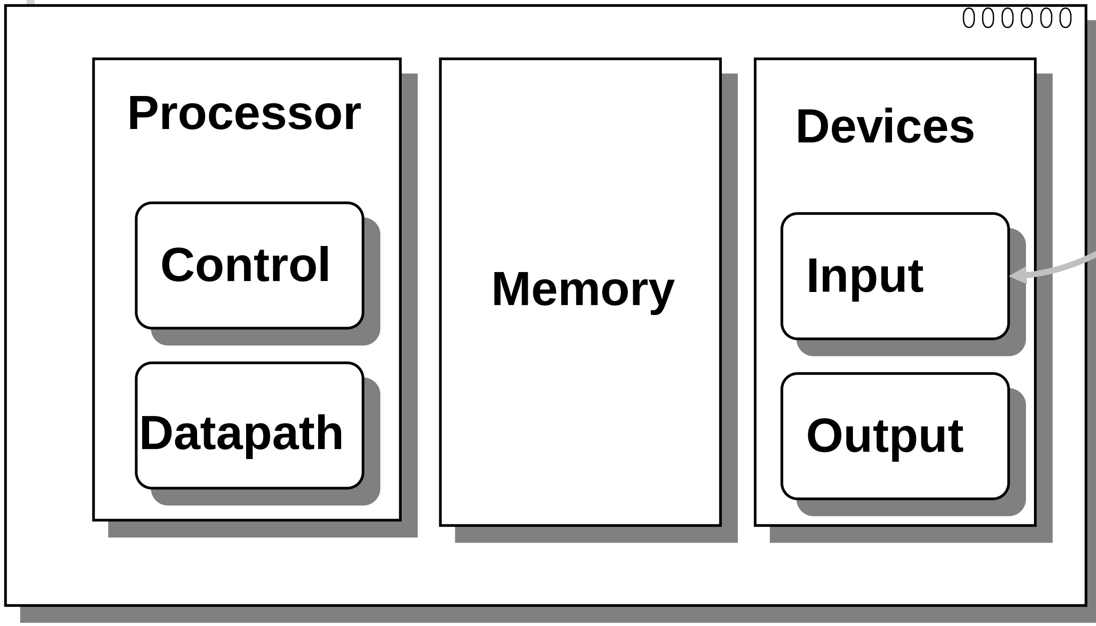
Major Components of a Computer

Anatomy: 5 components of any Computer



Input Device Inputs Object Code

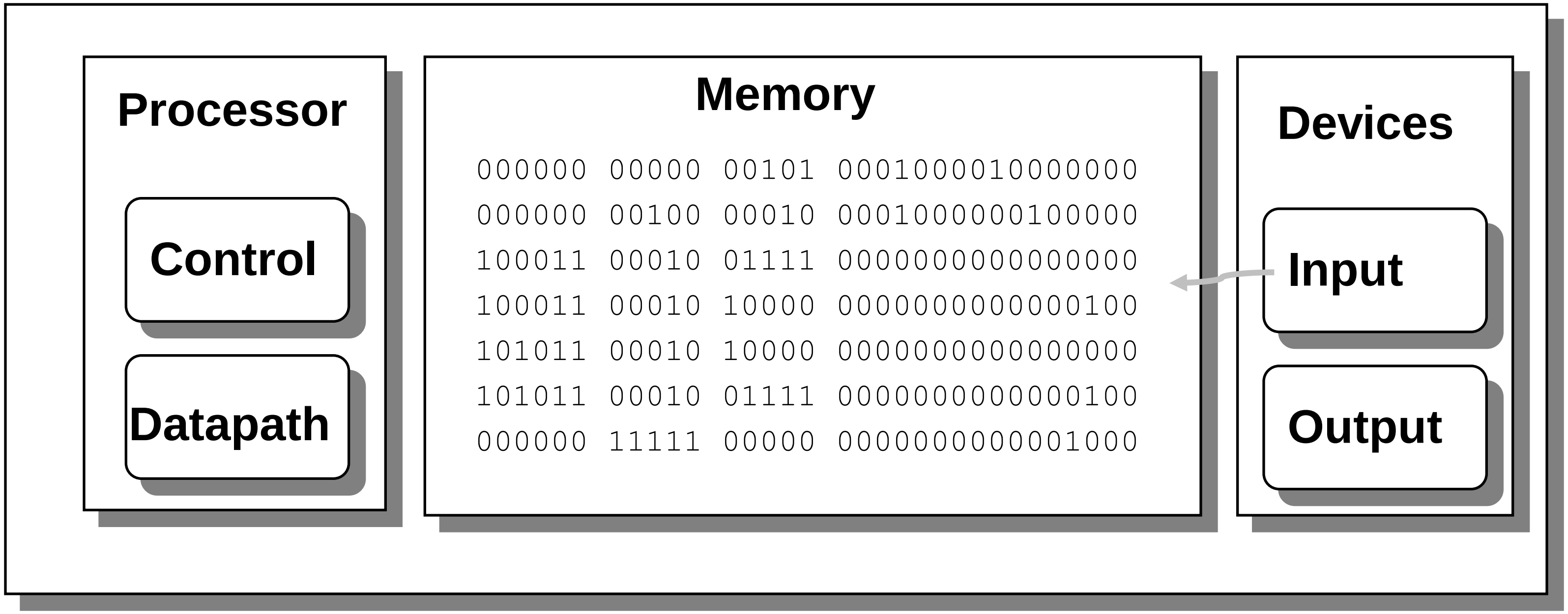
```
000000 00000 00101 0001000010000000
000000 00100 00010 0001000000100000
100011 00010 01111 0000000000000000
100011 00010 10000 0000000000000100
101011 00010 10000 0000000000000000
101011 00010 01111 0000000000000100
000000 11111 00000 0000000000001000
```



- Input devices
  - Keyboard
  - Mouse
  - Network
  - Joysticks, trackballs, etc

Input devices bring the object code and input data from the outside world into computer.

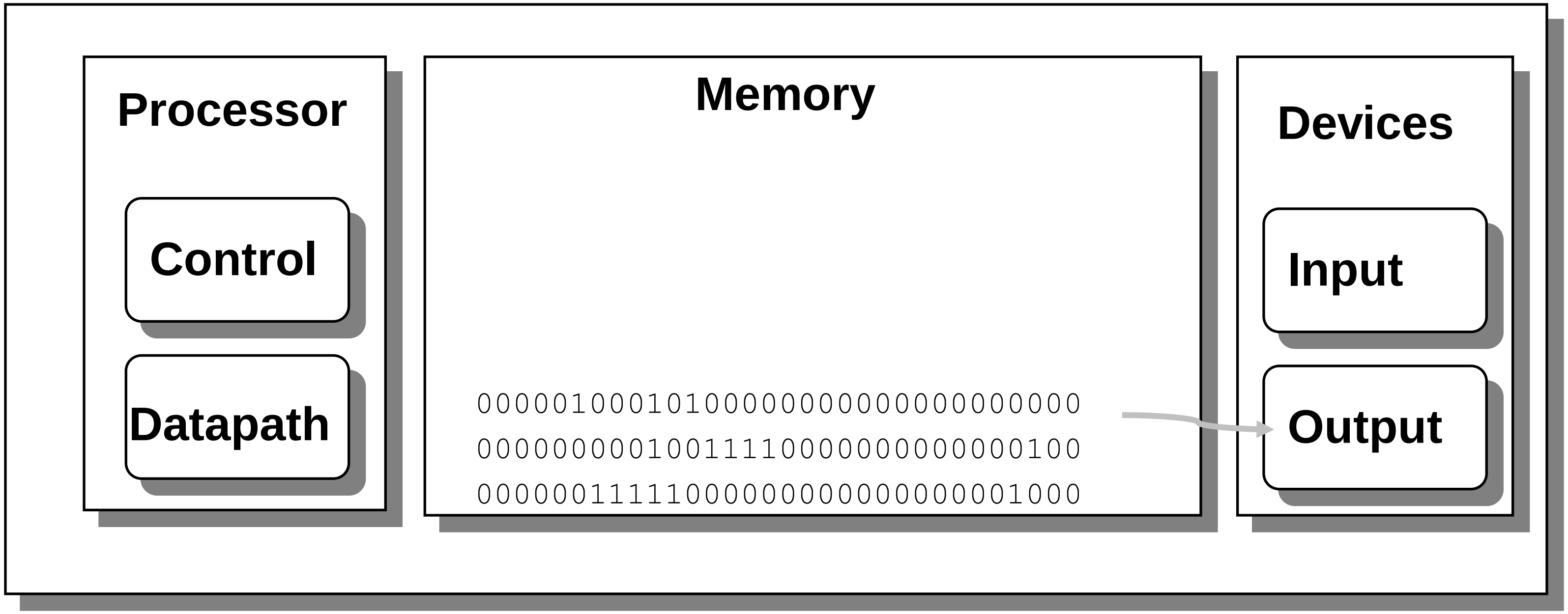
Object Code Stored in Memory



Memory holds both INSTRUCTIONS and DATA and you can't tell the difference. They are both just 32 bit strings of zeros and ones.

Output Data Stored in Memory

At program completion the data to be output resides in memory



Output Device Outputs Data